managing the process

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GR5069: Applied Data Science
for Social Scientists

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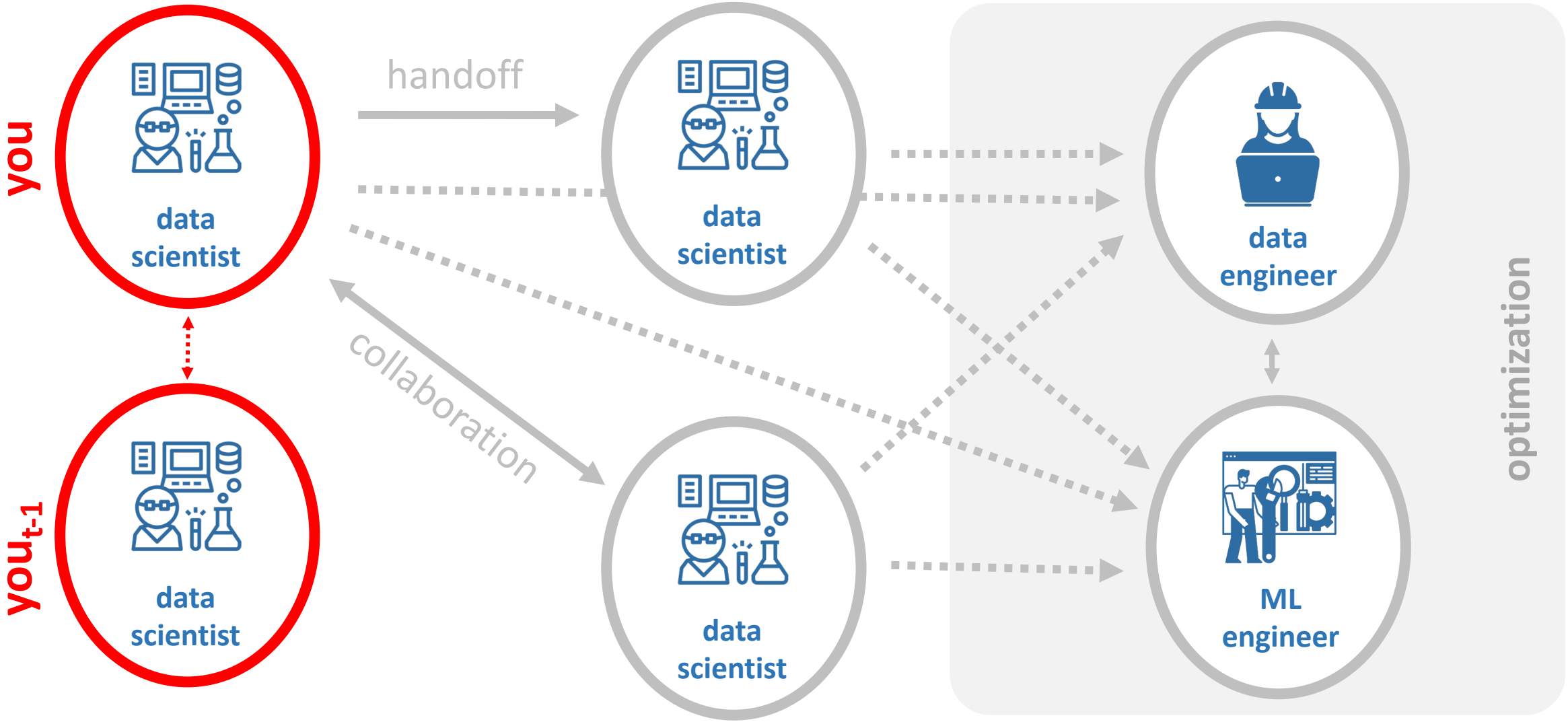
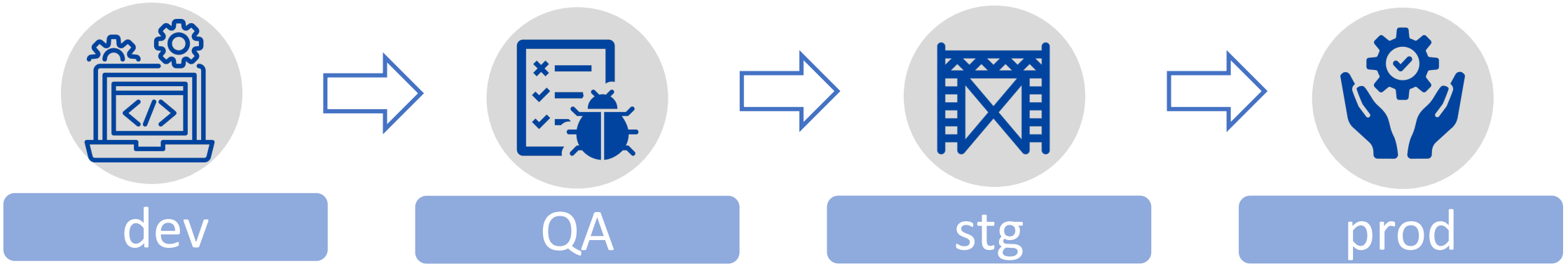
Columbia University

housekeeping

recap: workflow collaboration in Data Science



recap: working environments



development

quality assurance / testing

staging

production

recap: operational concepts in Data Science

portability

anyone should be able to **pick up where you left off** from any machine

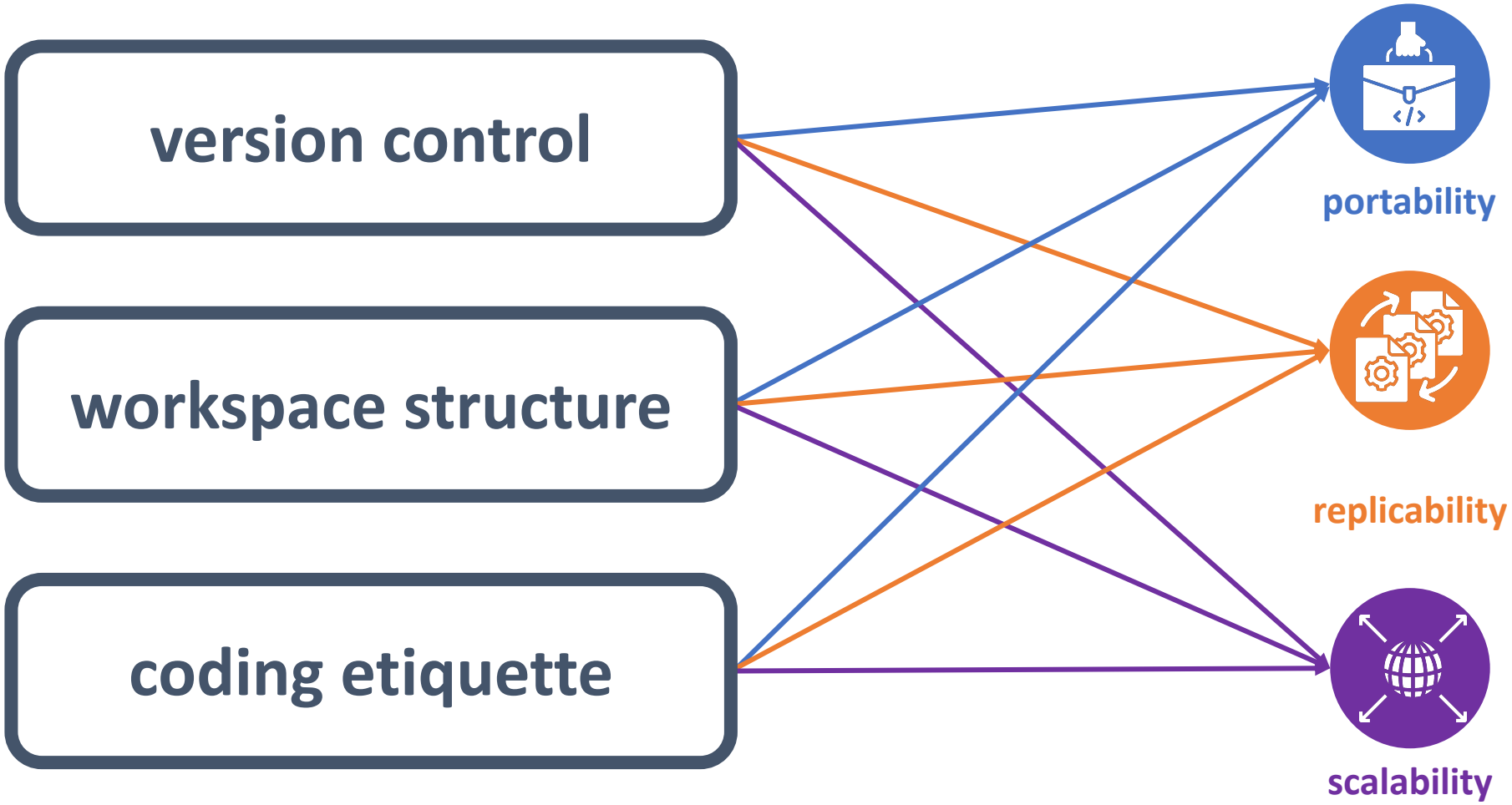
replicability

anyone should be able to arrive at your **same results**

scalability

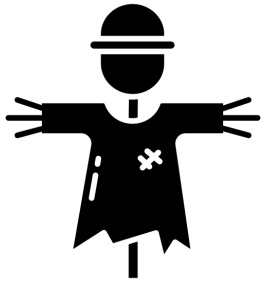
your prototype should also work for **larger data sets** and/or be on the path of **automation**

all pieces to support operational concepts

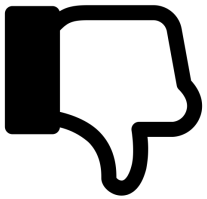


our focus today:

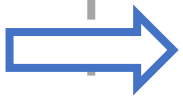
recap: iteration to build Data Products

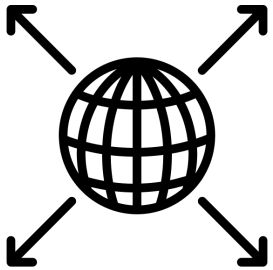


start small
(MVP)

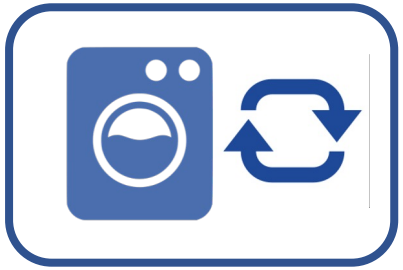


fail fast





scale up



iterate

a tale of **two approaches**…

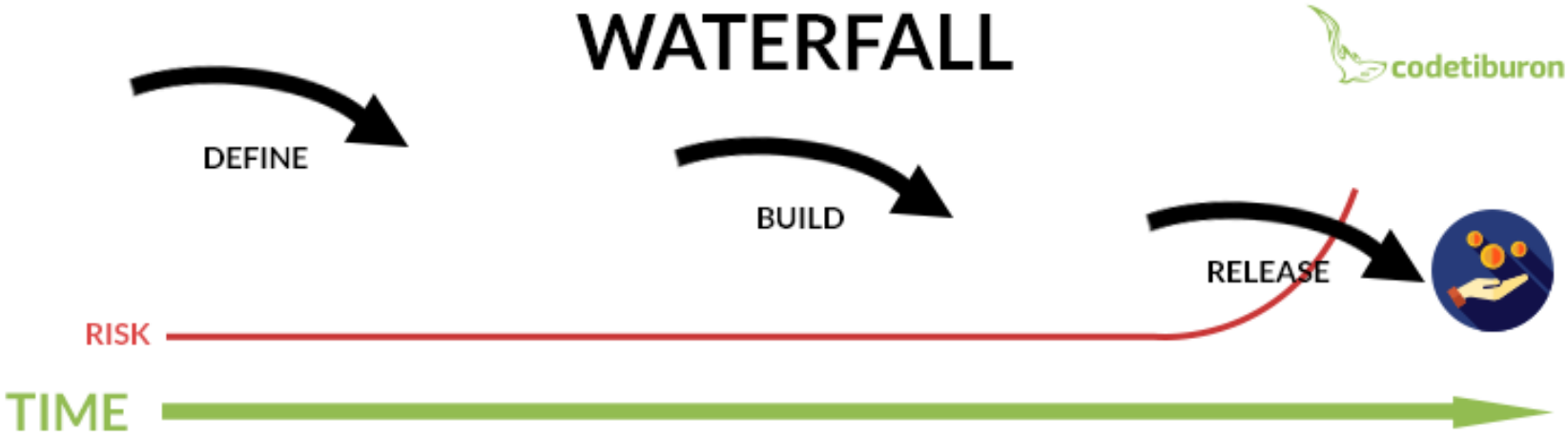


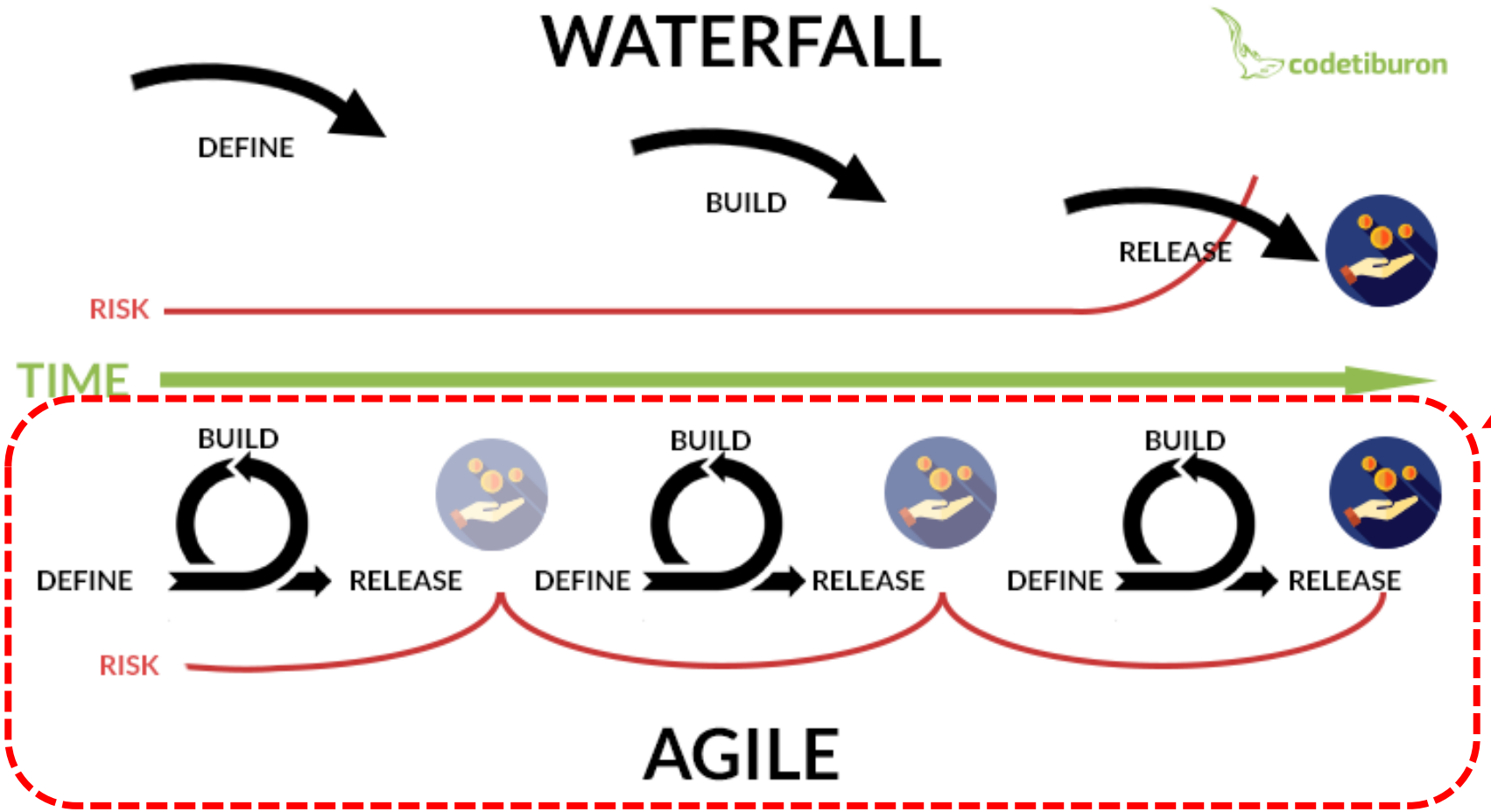
figure: [codetiburon]

definition

for our purposes, **waterfall** is

- **sequential** approach to manage **development processes**
  - where completing a stage is a **prerequisite** to start the next one
- aiming to release a **single final product**

a tale of **two approaches**…



our focus today

figure: [codetiburon]

definition

for our purposes, **Agile** is

iterative approach to manage **development processes**

focused on **continuous releases**

that incorporate **adjustments** at every iteration based on **user feedback**

enabling **planning for unplanned work**

Agile as a **dynamic** and **user-centric** approach

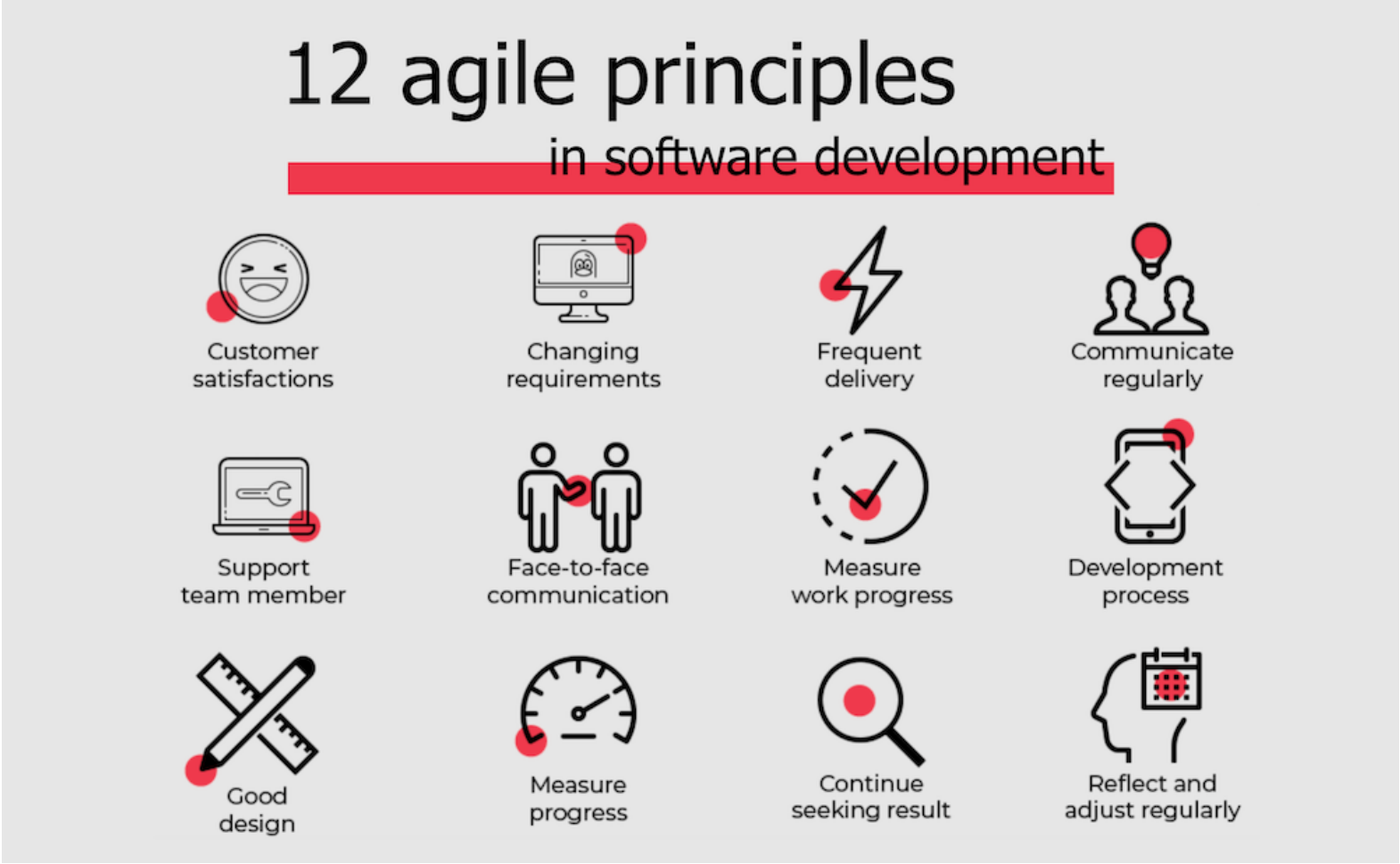
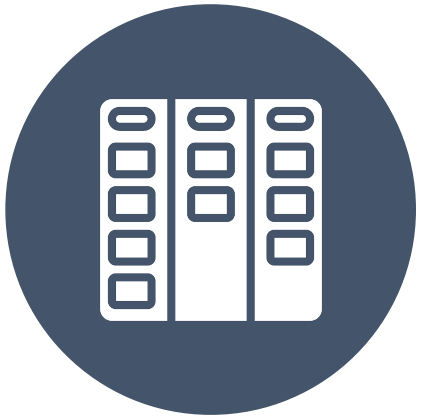


figure: [designveloper]

with a little help from my (Agile) friends!

scrum



Kanban

two commonly used **methods** to **implement** the Agile philosophy to **build** and **evolve** digital solutions

scrum as a structured method to be Agile

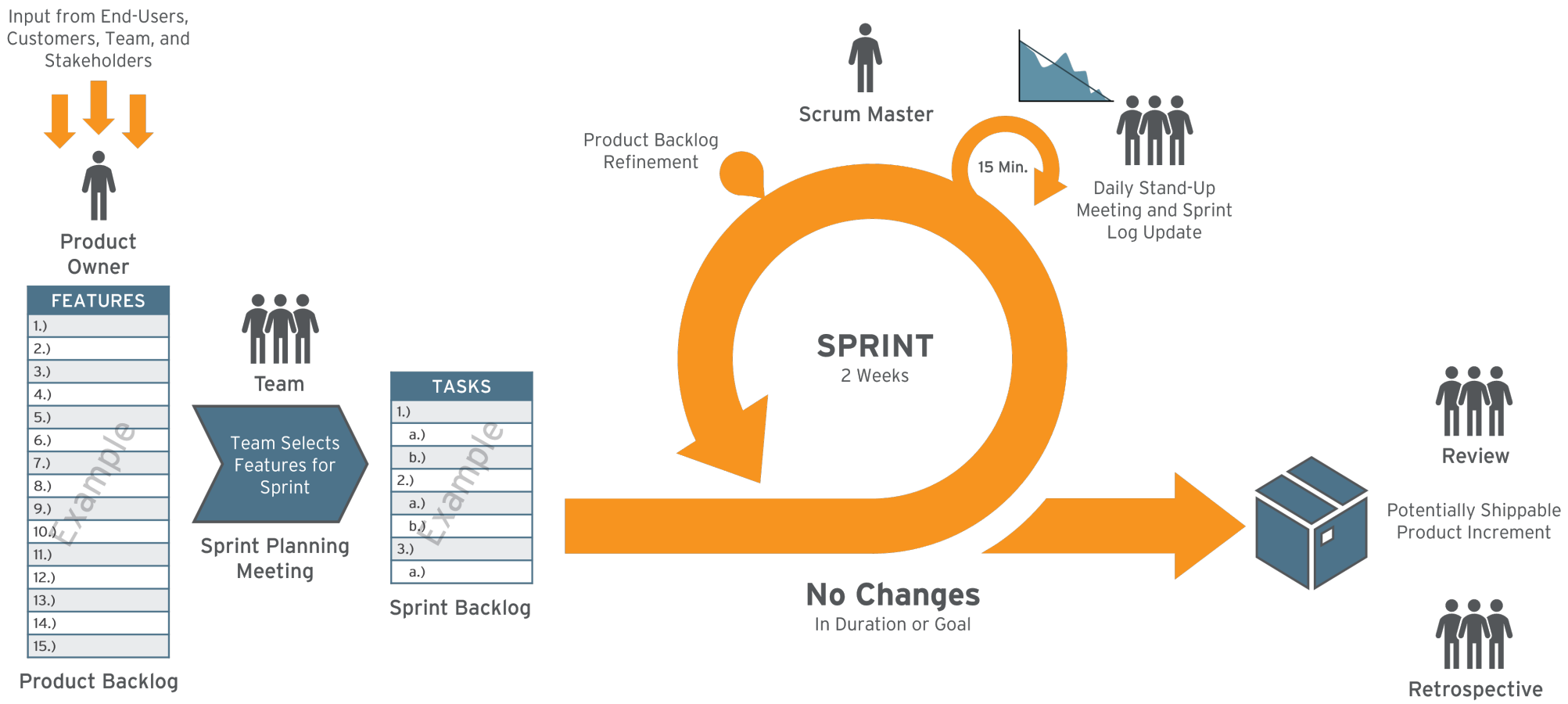


figure: clientek

for everything else, there’s **kanban** cards

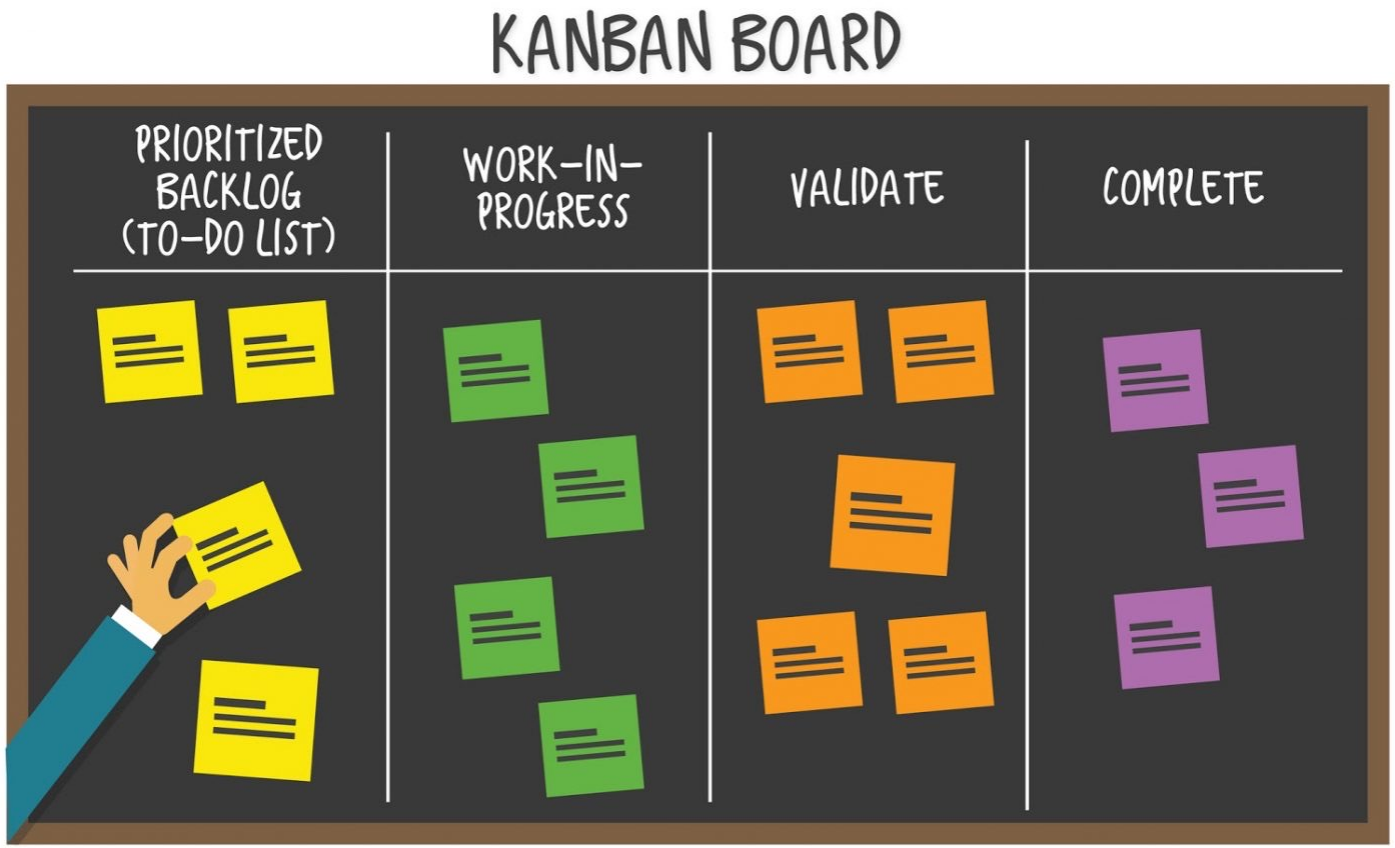
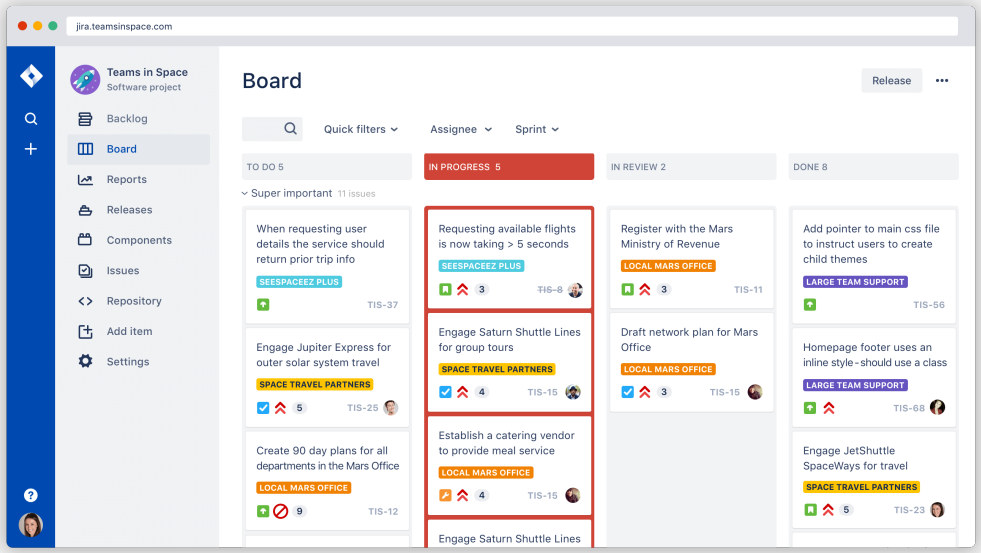
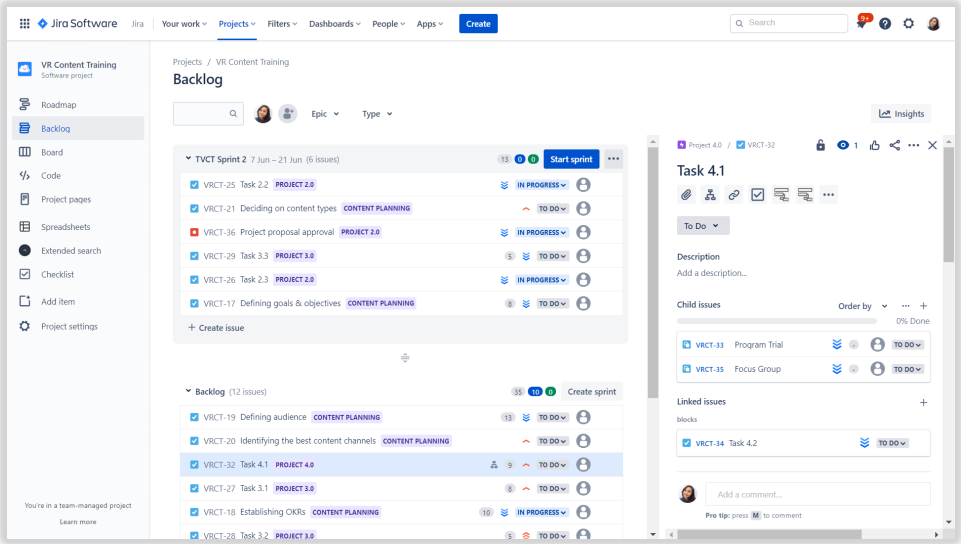


figure: [talentvis]

plenty of (software) **tools** to be Agile



Kanban board



SCRUM backlog

asana

ATLASSIAN Jira

Trello

a key **concept** to grasp: **MVPs**!

definition

for our purposes, an MVP is

- a **version** of a product
- with just **enough features** to be usable
- that enables collecting **feedback**
- with **minimal effort**

important to clearly <u>understand</u> MVPs!

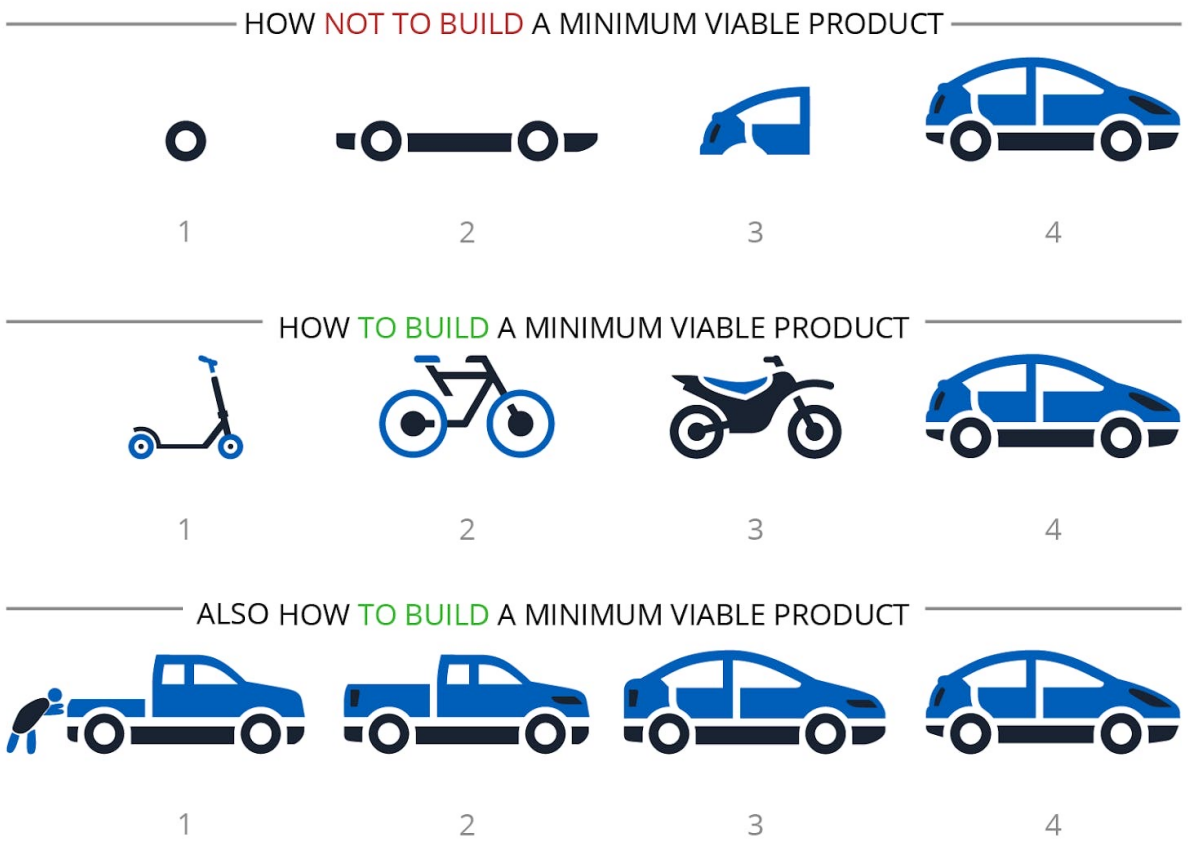
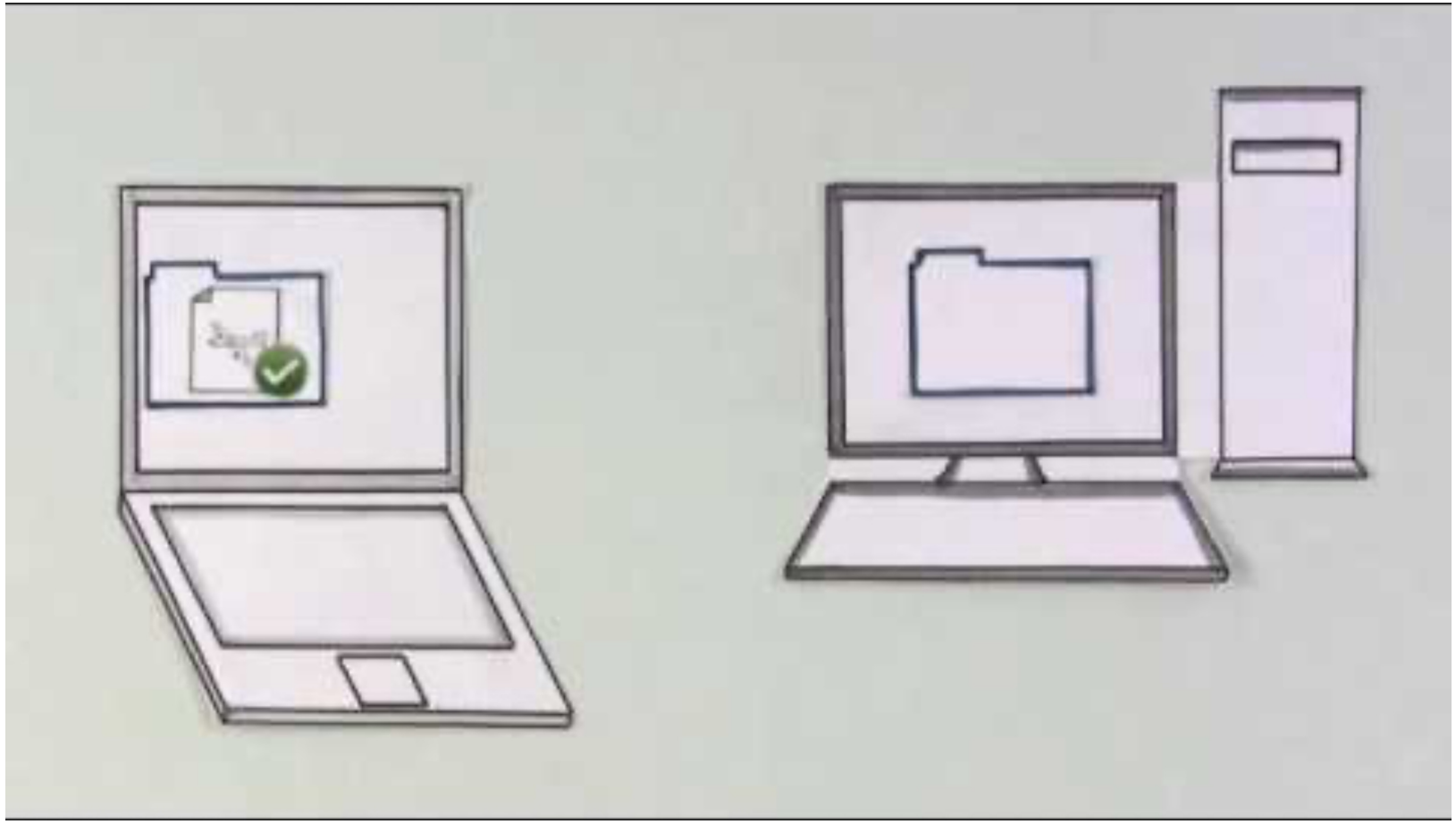


figure: altkom

building **MVPs** has useful **advantages**

- **safe** way to **validate** solution
- early **user feedback**
- lower **upfront costs**
- **easier iterative development**
- faster **time-to-market***

from the hall of fame of MVPs: **Dropbox**



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